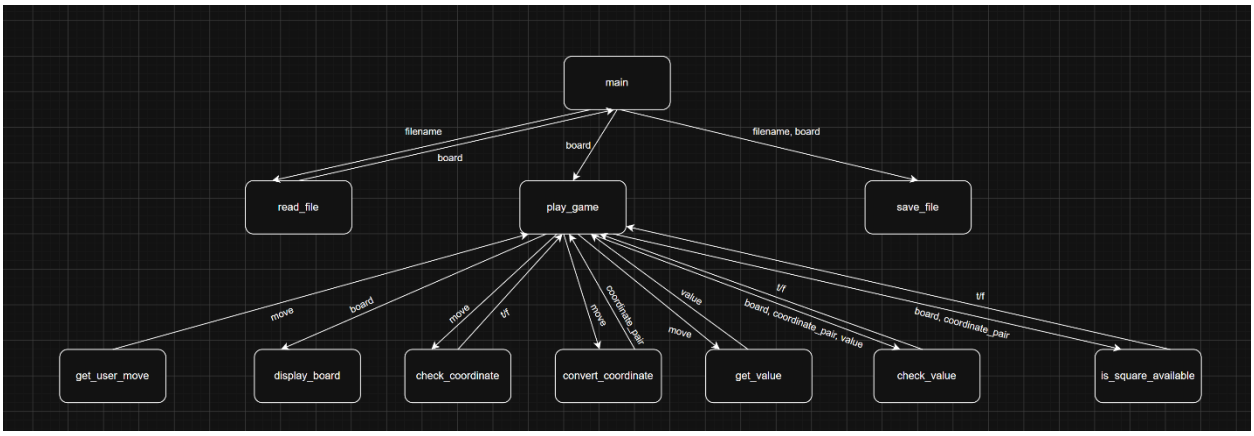
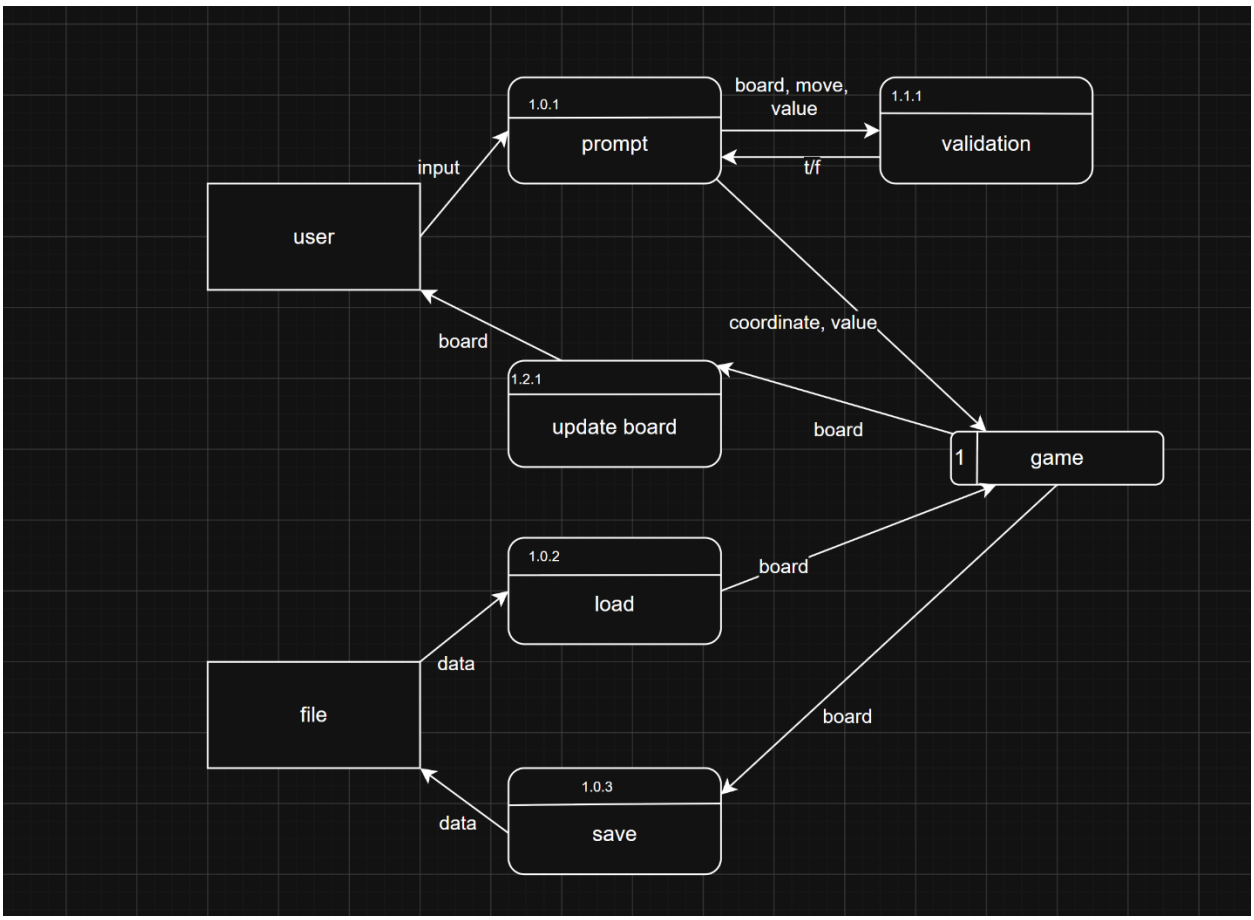
Step 1: Structure Chart



Step 2: Data Flow Diagram



Step 3 Pseudocode functions:

```
convert_coordinate(coordinate)
   column_map <- [A, B, C, D, E, F, G, H, I]
```

```
FOR column <- 0 … len(column_map) – 1
   IF column_map[column] == coordinate[0]
       column_num <- column
row_int <- int(coordinate[1])
row_num <- row_int - 1
coordinate_pair <- (column_num, row_num)
RETURN coordinate_pair
```

1. AI prompt for check_value function:

   I am creating a sudoku program in pseudocode that has the following constraints: There is no more than one instance of a given number on a given row.

- There is no more than one instance of a given number on a given column.
- There is no more than one instance of a given number on an inside square (the 3x3 squares embedded in the 9x9 grid).
- Every square can consist of a single digit between 1 and 9 exclusively or can be blank.

   Write me a function in pseudocode called check_value that determines if a given number can legally go in a given square in a Sudoku board. The function will receive board, coordinate_pair, and value as parameters. board is a 2d list with column headers A-I and row headers 1-9. coordinate_pair is a tuple like (1,4) that represents (column, row) to fill. value is an integer between 1 and 9 representing the value number that will fill a square.

2. AI-generated pseudocode:

```
check_value(board, coordinate_pair, value)
  column <- coordinate_pair[0]
  row <- coordinate_pair[1]

  IF value < 1 OR value > 9
    RETURN FALSE

  IF board[row][column] != 0
    RETURN FALSE
```

```
  # Check for duplicates in the selected row
  FOR col <- 0 ... 8
    IF board[row][col] == value
      RETURN FALSE

  # Check for duplicates in the selected column
  FOR r <- 0 ... 8
    IF board[r][column] == value
      RETURN FALSE

  # Find the top-left corner of the 3x3 square
  box_row <- (row DIV 3) * 3
  box_col <- (column DIV 3) * 3

  # Check for duplicates inside the 3x3 square
  FOR r <- box_row ... box_row + 2
    FOR col <- box_col ... box_col + 2
      IF board[r][col] == value
        RETURN FALSE

  # No Sudoku rules were violated
  RETURN TRUE
END
```

3. Changes to AI-generated pseudocode:

   I already have an is_square_available() function that checks if a square is occupied. So, I can remove that check in the check_value function.

```
check_value(board, coordinate_pair, value)
   column <- coordinate_pair[0]
   row <- coordinate_pair[1]
   IF value < 1 OR value > 9
      RETURN FALSE
   # Check for duplicates in the selected row
   FOR col <- 0 ... 8
      IF board[row][col] == value
         RETURN FALSE

   # Check for duplicates in the selected column
```

```
   FOR r <- 0 ... 8
      IF board[r][column] == value
         RETURN FALSE
   # Find the top-left corner of the 3x3 square
   box_row <- (row DIV 3) * 3
   box_col <- (column DIV 3) * 3
   # Check for duplicates inside the 3x3 square
   FOR r <- box_row ... box_row + 2
      FOR col <- box_col ... box_col + 2
         IF board[r][col] == value
            RETURN FALSE
   # No Sudoku rules were violated
   RETURN TRUE
END
```